# DATASCI 241: Problem Set 3

David Schaaf

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# 1 Peruvian Recycling

Look at the article located here (./readings/recycling_peru.pdf) about encouraging recycling in Peru. The paper contains two experiments, a "participation study" and a "participation intensity study." In this problem, we will focus on the latter study, whose results are contained in Table 4 in this problem. You will need to read the relevant section of the paper (starting on page 20 of the manuscript) in order to understand the experimental design and variables. (*Note that "indicator variable" is a synonym for "dummy variable," in case you have not seen this language before.*)

## 1.1 Recycling bin ATE

In Column 3 of Table 4A, what is the estimated ATE of providing a recycling bin on the average weight of recyclables turned in per household per week, during the six-week treatment period? Provide a 95% confidence interval. Provide a short narrative using inline R code, such as r inline_reference.

```
# Avg. weight (in kg) of recyclables turned in per week

any_bin <- 0.187
any_bin_se <- 0.032
n <- 1781
k <- 4
deg_f <- n - k - 1
any_bin_critical_t <- qt(0.975, deg_f)
any_bin_range_1_side <- any_bin_critical_t * any_bin_se
any_bin_lower_bound <- round(any_bin - any_bin_range_1_side, 3)
any_bin_upper_bound <- round(any_bin + any_bin_range_1_side, 3)
```

**Answer:**

The average treatment effect (ATE) for any recycling bin in terms of kg of recyclables turned in per week is 0.187 kg/week. The standard error for this coefficient is 0.032. The 95% confidence interval (degrees of freedom = 1776) for the average treatment effect is [0.124, 0.25] kg/week.

## 1.2 SMS ATE

In Column 3 of Table 4A, what is the estimated ATE of sending a text message reminder on the average weight of recyclables turned in per household per week? Provide a 95% confidence interval and provide a short narrative using inline R code.

```
any_sms = -0.024
any_sms_se = 0.039

any_sms_critical_t <- qt(0.975, deg_f)
any_sms_range_1_side <- any_sms_critical_t * any_sms_se
any_sms_lower_bound <- round(any_sms - any_sms_range_1_side, 3)
any_sms_upper_bound <- round(any_sms + any_sms_range_1_side, 3)
```

**Answer:** The average treatment effect (ATE) for any SMS message in terms of kg of recyclables turned in per week is -0.024 kg/week. The standard error for this coefficient is 0.039. The 95% confidence interval (degrees of freedom = 1776) for the average treatment effect is [-0.1, 0.052] kg/week.

## 1.3 What outcomes does a recycling bin affect?

Which outcome measures in Table 4A show statistically significant effects (at the 5% level) of providing a recycling bin? How are you dealing with the issue that there are several different tests that have been run, and that you are reading? How, if at all, do the authors deal with this?

**Answer:**

In Table 4A, the any bin coefficients are significant at the 1% level for all outcomes except for the contamination outcome. While they make 5 tests of the any bin coefficient, the authors do not make an adjustment to the significance threshold for the multiple tests. These are 5 different outcomes of the same treatment (providing a recycling bin), that show the treatment had the intended effect. They are not close to the borderline significance threshold of 5% either. This indicates that the treatment effect was strong enough that the impact can be measured in multiple different ways.

## 1.4 What outcomes does a SMS affect?

Which outcome measures in Table 4A show statistically significant effects (at the 5% level) of sending text messages? Now that you have read across two different treatments, and many outcomes, what, if anything do the p-values mean to you? Does this feel like p-hacking or doing careful investigation?

**Answer:**

None of the outcomes show significance for SMS messaging treatment. The authors discuss that the tests "do not indicate a clear impact of SMS reminders", and they are dismissive of the SMS messaging in their final conclusions. The authors control for the cell phone indicator, which does have some measurable effect in outcomes (2), (3), and (4) at the 5% level. This is not a treatment condition, so it does not have causal effect. Rather it serves as a reminder that the SMS messaging is not a purely random treatment as it first requires that the household has a cell phone. This does not feel like p-hacking as the authors have considered pre-treatment covariates with their study. By controlling for this covariate, the authors avoid attributing the variation of the cell phone population to their treatment.

## 1.5 Marginal effects

Suppose that, during the two weeks before treatment, household A turns in 2kg per week more recyclables than household B does, and suppose that both households are otherwise identical (including being in the same treatment group). From the model, how much more recycling do we predict household A to have than household B, per week, during the six weeks of treatment? Provide only a point estimate, as the confidence interval would be a bit complicated. This question is designed to test your understanding of slope coefficients in regression.

```
h_diff = 2 # kg per week

baseline <- 0.281
baseline_se <- 0.011

h_diff_treat = h_diff * baseline
```

**Answer:**

The coefficient on baseline in model (3) of 0.281 means that 1kg of baseline recycling leads to 0.281 kg more recycling during the treatment period. If household A turns in 2kg more per week than household B, then during the treatment period we would expect household A to turn in 2 x 0.281 = 0.562 kg per week more than household B.

## 1.6 Covariates or confounders?

Suppose that the variable "percentage of visits turned in bag, baseline" had been left out of the regression reported in Column 1. What would you expect to happen to the results on providing a recycling bin? Would you expect an increase or decrease in the estimated ATE? Would you expect an increase or decrease in the standard error? Explain your reasoning.

```
# Use this code chunk to show your code work (if needed)
```

**Answer:**

The baseline tells us how much a household's pre-treatment recycling habits impacted their treatment period recycling. The coefficient on "percentage of visits turned in bag, baseline" is positive, so most households recycled more during the treatment period whether they received the treatment or not. If this was removed then we risk attributing that variation to the recycling bin treatment, which would increase the coefficient and overstate the treatment effect. This would also increase the standard error of the coefficient as there is more unexplained variance in the model.

## 1.7 Bad control or useful subset?

In column 1 of Table 4A, would you say the variable "has cell phone" is a bad control? Explain your reasoning, and engage both with the definition of a bad control, and also the implications of including a bad control in a model.

**Answer:**

We add control variables to account for bias in our model that would interfere with drawing the proper conclusion from the analysis. A bad control can have the opposite impact and add bias to the model by soaking up some of the measurable impact of the treatment effect. In this case, the "has cell phone" coefficient is significant, but the SMS messaging coefficient is not. So we must question if this control variable is taking credit for the treatment effect, or is removing bias from our model.

The authors identify that cell phone users are "slightly richer, more educated, and more interested in local affairs (especially recycling matters) than non-cell phone owners". "Has cell phone" is a pre-treatment covariate, rather than a post-treatment covariate. There is a systematic difference between cell phone users and non-cell phone users. To receive the SMS treatment, using a cell phone is a pre-requisite, so the treatment is not truly randomly assigned. The "has cell phone" control variable is recognition that the SMS treatment is not fully random, and they need to account for how this group systematically differs from people who were ineligible from receiving the treatment. Given the background information on the adoption of cell phones in Peru, this is not a bad control. This variable defines the difference between a class of people who were eligible for the SMS treatment and those who were not.

## 1.8 What happens if you remove "has cell phone"?

If we were to remove the "has cell phone" variable from the regression, what would you expect to happen to the coefficient on "Any SMS message"? Would it go up or down? Explain your reasoning.

```
# Use this code chunk to show your code work (if needed)
```

**Answer:**

If we remove "has cell phone" from the regression, the coefficient on the "Any SMS message" would increase. The coefficients for this control variable are positive and greater in magnitude than the treatment variable across the board. The SMS message variable would be more positive because the cell phone users generally recycle more than non-cell phone users. It would not be fair to compare people who both have a cellphone and received an SMS message to people who do not own a phone and could never have been randomly assigned into the treatment.

# 2 Multifactor Experiments

Continue to use the same experiment setup from problem 1. Now think about multifactor experiments.

## 2.1 Experiment design?

What is the full experimental design for this experiment? Tell us the dimensions, such as 2x2x3. The full results appear in Panel 4B. We'll note that the dimensions of an experiment are defined in terms of the *treatments that the experiment assigns*, not in terms of other features about the data.

**Answer:**

The experiment assign a treatment condition for Bins with 3 conditions: No bin, Bin without sticker, Bin with sticker, and a treatment condition for SMS with 3 conditions: No SMS, Personal message SMS, Generic SMS. This is a 3x3 experiment design.

## 2.2 Baseline for interpretation

In the results of Table 4B, describe the baseline category. That is, in English, how would you describe the attributes of the group of people for whom all dummy variables are equal to zero?

**Answer:**

The baseline category represented no bin and no cell phone and therefore also no SMS. This group is the baseline for all treatment conditions and the baseline for the "has cell phone" variable too. This represents the behaviors of someone who is unaffected by any treatment condition who on average also fits the profile of the class that does not own cell phones (less wealthy, less concerned about local issues like recycling).

## 2.3 Bin without sticker effect

In column (1) of Table 4B, interpret the magnitude of the coefficient on "bin without sticker." What does it mean?

**Answer:**

Columns (1) represents the percentage of visits that the household turned in a recycling bag or bin when they had the opportunity to. The coefficient is interpreted as percentage points, so the coefficient of 0.035 on "bin without sticker" is interpreted as an increase of 3.5 percentage points in in the frequency the household turned in a bag compared to baseline.

## 2.4 With or without a sticker?

In column (1) of Table 4B, which seems to have a stronger treatment effect, the recycling bin with message sticker, or the recycling bin without sticker? How large is the magnitude of the estimated difference?

**Answer:**

In column (1), the "bin with sticker" has treatment effect of 5.5 percentage points compared to "bin without sticker" at 3.5 percentage points. Both are significant, but the "bin with sticker" is significant at the 1% level while the "bin without sticker" is significant at the 5% level. Looking solely at these 2 numbers, the "bin with sticker" has a stronger treatment effect by 2.0 percentage points.

## 2.5 Statistical significantly different with or without a sticker?

Is this difference you just described statistically significant? Explain which piece of information in the table allows you to answer this question.

**Answer:**

The authors provide an F-test p-value that (1) bin with sticker = (2) bin without sticker. This test has a value of 0.31 in column (1), so this is not a statistically significant difference.

## 2.6 Fully saturated?

Notice that Table 4C is described as results from "fully saturated" models. What does this mean? What does David Reiley propose this definition means to him in the async lecture? What do the authors seem to think it means to them? Looking at the list of variables in the table, explain in what sense the model is "saturated."

**Answer:**

A fully saturated model includes coefficients for all measurable variables and all possible interaction effects. David Reiley specifies that each value of the covariate and every possible interaction gets a different dummy variable to represent it, including every value of continuous variables (e.g. SAT scores 1400, 1410, 1420 etc. each have their own dummy variable). With a different coefficient on every possible variable, the model is non-linear for continuous variables because we have a point estimate for every individual value.

The authors present a fully saturated model in Table 4C. This model includes dummy variables for all possible treatment interactions of the Bin and SMS messaging treatments. It also includes a dimension for the "has cell phone" vs "no phone" groups, and a variable "has cell phone" which would change the baseline for the group in the SMS experiment. While not an experimental treatment, it does create another split in the data. Notably, their version of a fully saturated model does not include dummy variables and interactions with the baseline measurement in the pre-treatment phase, which remains as a continuous variable.

| Bin Treatment with Has cell phone | No SMS | Generic | Personal |
|---|---|---|---|
| **No Bin** | yes | yes | yes |
| **Bin without sticker** | yes | yes | yes |
| **Bin with sticker** | yes | yes | yes |

| Bin Treatment with No phone | No Cell Phone |
|---|---|
| **No Bin** | Baseline |
| **Bin without sticker** | yes |
| **Bin with sticker** | yes |

# 3 Now! Do it with data

Download the data set for the recycling study in the previous problem, obtained from the authors. We will be focusing on the outcome variable Y="number of bins turned in per week" (avg_bins_treat).

## 3.1 Treatment only model

A. For simplicity, let's start by measuring the effect of providing a recycling bin, ignoring the SMS message treatment (and ignoring whether there was a sticker on the bin or not). Run a regression of Y on only the bin treatment dummy, so you estimate a simple difference in means. Provide a 95% confidence interval for the treatment effect, using **of course** robust standard errors (use these throughout) and provide a brief narrative using R inline statements.

```
mod_basic <- lm(avg_bins_treat ~ bin , data = d)
mod_basic_rse <- sqrt(diag(vcovHC(mod_basic)))

mod_basic_ci <- ci_95(mod_basic, 'bin')

mod_basic_bin_coef <- mod_basic_ci$coef["bin"]
mod_basic_bin_rse <- mod_basic_ci$se['bin']
mod_basic_bin_lower <- mod_basic_ci$lower['bin']
mod_basic_bin_upper <- mod_basic_ci$upper['bin']

stargazer(mod_basic,
          se = list(mod_basic_rse),
          covariate.labels = c("Any bin", "Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

Table 3:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Any bin | 0.135*** |
| | (0.021) |
| Constant | 0.635*** |
| | (0.011) |
| Observations | 1,785 |
| $R^2$ | 0.024 |
| Adjusted $R^2$ | 0.024 |
| Residual Std. Error | 0.405 (df = 1783) |
| F Statistic | 44.516*** (df = 1; 1783) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

**Narrative:**

The basic model regresses the average number of bins turned in on the bin treatment indicator with no controls. The coefficient for the bin treatment is 0.135 with a heteroskedastic consistent standard error of 0.021. This is significant at the 1% level with a 95% confidence interval of [0.095, 0.176].

## 3.2 Treatment and pre-treatment values

Now add the pre-treatment value of Y as a covariate. Provide a 95% confidence interval for the treatment effect. Explain how and why this confidence interval differs from the previous one.

```
mod_pretreat <- lm(avg_bins_treat ~ bin + base_avg_bins_treat, data = d)
mod_pretreat_rse <- sqrt(diag(vcovHC(mod_pretreat)))

mod_pretreat_ci <- ci_95(mod_pretreat, 'bin')

mod_pretreat_bin_coef <- mod_pretreat_ci$coef["bin"]
mod_pretreat_bin_rse <- mod_pretreat_ci$se['bin']
mod_pretreat_bin_lower <- mod_pretreat_ci$lower['bin']
mod_pretreat_bin_upper <- mod_pretreat_ci$upper['bin']
mod_pretreat_r2 <- r2 <- round(summary(mod_pretreat)$r.squared, 3)
stargazer(mod_pretreat,
          se = list(mod_pretreat_rse),
          covariate.labels = c("Any bin", "Pre-treat Baseline","Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

Table 4:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Any bin | 0.124*** |
| | (0.017) |
| Pre-treat Baseline | 0.390*** |
| | (0.031) |
| Constant | 0.352*** |
| | (0.021) |
| Observations | 1,782 |
| $R^2$ | 0.338 |
| Adjusted $R^2$ | 0.337 |
| Residual Std. Error | 0.333 (df = 1779) |
| F Statistic | 453.518*** (df = 2; 1779) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

**Answer:**

The pre-treatment adjusted model regresses the average number of bins turned in on the bin treatment indicator with a control for the households baseline number of bins turn per week in pre-treatment. The coefficient for the bin treatment decreases from the basic model to 0.124 with a heteroskedastic consistent standard error of 0.017, which is also less than the basic model's standard error. This is significant at the 1% level with a 95% confidence interval of [0.091, 0.158].

## 3.3 Add street fixed effects

Now add the street fixed effects. (You'll need to use the R command `factor()` You can do this either within the `lm` call, or you can move this factoring up in the data pipeline so that it persists through the rest of your

analysis. The only thing we would recommend that you *not* do is to engineer a new, persistent feature at this point.) Provide a 95% confidence interval for the treatment effect and provide a brief narrative using r inline statements.

```
mod_fixed_effects <- lm(avg_bins_treat ~ bin +
                          base_avg_bins_treat + street, data = d)

mod_fe_se <- sqrt(diag(vcovHC(mod_fixed_effects)))

mod_fe_ci <- ci_95(mod_fixed_effects, 'bin')

mod_fe_bin_coef <- mod_fe_ci$coef["bin"]
mod_fe_bin_rse <- mod_fe_ci$se['bin']
mod_fe_bin_lower <- mod_fe_ci$lower['bin']
mod_fe_bin_upper <- mod_fe_ci$upper['bin']
mod_fe_r2 <- r2 <- round(summary(mod_fixed_effects)$r.squared, 3)
stargazer(mod_fixed_effects,
          se = list(mod_fe_se),
          omit = "street",
          omit.labels = "Street Fixed Effects",
          covariate.labels = c("Any bin", "Pre-treat Baseline", "Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

Table 5:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Any bin | 0.114*** |
| | (0.019) |
| Pre-treat Baseline | 0.374*** |
| | (0.030) |
| Constant | 0.368*** |
| | (0.035) |
| Street Fixed Effects | Yes |
| Observations | 1,782 |
| $R^2$ | 0.436 |
| Adjusted $R^2$ | 0.372 |
| Residual Std. Error | 0.324 (df = 1600) |
| F Statistic | 6.840*** (df = 181; 1600) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

```
mod_fixed_effects$coefficients["(Intercept)"]
```

(Intercept) 0.367744

**Narrative:**

The model now includes the household baseline and fixed effects for the street. The coefficient for the bin treatment decreases again to 0.114 with a heteroskedastic consistent standard error of 0.019, which has

increased from the model with baseline. This is still significant at the 1% level with a 95% confidence interval of [0.077, 0.151].

### 3.4 Test for block fixed effects

Recall that the authors described their experiment as "stratified at the street level," which is a synonym for blocking by street. Does including these block fixed effects change the standard errors of the estimates *very much*? Conduct the appropriate test for the inclusion of these block fixed effects, and interpret them in the context of the other variables in the regression.

```
test_fixed_effects <- anova(mod_pretreat, mod_fixed_effects, test = "F")
test_fe_f <- test_fixed_effects$F[2]
test_fe_p <- test_fixed_effects$`Pr(>F)`[2]

change_se <- percent((mod_fe_bin_rse - mod_pretreat_bin_rse)/mod_pretreat_bin_rse)
test_fixed_effects
```

```
## Analysis of Variance Table
##
## Model 1: avg_bins_treat ~ bin + base_avg_bins_treat
## Model 2: avg_bins_treat ~ bin + base_avg_bins_treat + street
##   Res.Df    RSS  Df Sum of Sq      F    Pr(>F)
## 1   1779 196.78
## 2   1600 167.50 179    29.278 1.5624 9.514e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Answer:**

The baseline model without street fixed effects has a standard error of 0.017 which increase 12% to 0.019 after the fixed effects are included. I performed an F test comparing the pre-treatment model and the pre-treatment with street fixed effects model. The F statistic is significant at the 0.1% level. Inclusion of the 179 new variables reduces the degrees of freedom from 1779 to 1600 which changes the calculation of variance enough to increase the standard error on the bin coefficient, while simultaneously increasing R-squared from 0.338 to 0.436. The street fixed effects explain more of the variance in the model than the baseline alone model, but it does not improve the precision of the estimate on the bin treatment.

### 3.5 Feature (no) cell phone

Perhaps having a cell phone helps explain the level of recycling behavior. Instead of "has cell phone," we find it easier to interpret the coefficient if we define the variable " no cell phone." Give the R command to define this new variable, which equals one minus the "has cell phone" variable in the authors' data set. Use "no cell phone" instead of "has cell phone" in subsequent regressions with this dataset.

Now add "no cell phone" as a covariate to the previous regression. Provide a 95% confidence interval for the treatment effect. Explain why this confidence interval does not differ much from the previous one.

```
mod_cellphone <- lm(avg_bins_treat ~ bin + no_cell_phone +
                      base_avg_bins_treat + street, data = d)

mod_cell_rse <- sqrt(diag(vcovHC(mod_cellphone)))

mod_cell_ci <- ci_95(mod_cellphone, 'bin')

mod_cell_coef <- mod_cell_ci$coef["bin"]
mod_cell_coef_rse <- mod_cell_ci$se['bin']
mod_cell_lower <- mod_cell_ci$lower['bin']
```

```r
mod_cell_upper <- mod_cell_ci$upper['bin']

mod_cell_const <- mod_cellphone$coefficients["(Intercept)"]
mod_cell_no_phone_const <- mod_cell_const + mod_cellphone$coefficients["no_cell_phone"]

stargazer(mod_cellphone,
          se = list(mod_cell_rse),
          omit = "street",
          omit.labels = "Street Fixed Effects",
          covariate.labels = c("Any bin", "No cell phone","Pre-treat Baseline",  "Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

Table 6:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Any bin | 0.115*** |
| | (0.019) |
| No cell phone | −0.050*** |
| | (0.018) |
| Pre-treat Baseline | 0.373*** |
| | (0.030) |
| Constant | 0.387*** |
| | (0.036) |
| Street Fixed Effects | Yes |
| Observations | 1,781 |
| $R^2$ | 0.439 |
| Adjusted $R^2$ | 0.375 |
| Residual Std. Error | 0.323 (df = 1598) |
| F Statistic | 6.875*** (df = 182; 1598) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

**Answer:**

We can define the new variable for no cell phone by `d[,'no_cell_phone'] <- 1 - d[,'havecell']`, which creates a new variable in the data table with `1` for no cell phone and `0` for has cell phone. This variable has been made at the top of the document as suggested.

The coefficient for the any bin treatment effect is 0.115 with a heteroskedastic consistent standard error of 0.019. This is significant at the 1% level with a 95% confidence interval of [0.078, 0.152].

The treatment effect of the bin remains similar to the previous model because the bin treatment was assigned independent of the cell phone status. If there were some correlation between cell phone and the treatment, even if incidental, the model would pick up on that and show a more substantial change in the confidence interval. Because this is not correlated with the treatment, we can view this variable as dividing the subjects into 2 segments based on this pre-treatment factor. In the previous model there the Constant term was 0.368 for all participants regardless of their cell phone status. By including this the "no cell phone" variable, the

Constant (0.387) of this model now reflects only those households *with* a cell phone. For the group with no cell phone we adjust this value by the coefficient on "no cell phone" to get a value of 0.338 before we consider the baseline or bin treatment.

### 3.6 Add the sms treatment

Now let's add in the SMS treatment. Re-run the previous regression with "any SMS" included. You should get the same results as in Table 4A. Provide a 95% confidence interval for the treatment effect of the recycling bin. Explain why this confidence interval does not differ much from the previous one.

```
mod_sms <- lm(avg_bins_treat ~ bin + sms + no_cell_phone +
                base_avg_bins_treat + street, data = d)

mod_sms_rse <- sqrt(diag(vcovHC(mod_sms)))

mod_sms_ci <- ci_95(mod_sms, 'bin')

mod_sms_coef <- mod_sms_ci$coef["bin"]
mod_sms_coef_rse <- mod_sms_ci$se['bin']
mod_sms_lower <- mod_sms_ci$lower['bin']
mod_sms_upper <- mod_sms_ci$upper['bin']

stargazer(mod_sms,
          se = list(mod_sms_rse),
          omit = "street",
          omit.labels = "Street Fixed Effects",
          covariate.labels = c("Any bin", "Any SMS", "No cell phone",
                               "Pre-treat Baseline", "Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

**Answer:**

The coefficient for the any bin treatment effect is 0.115 with a heteroskedastic consistent standard error of 0.019. This is significant at the 1% level with a 95% confidence interval of [0.078, 0.152].

This model includes the second treatment condition for SMS messaging, which is weak at 0.005 and not statistically significant. The treatment assignment of SMS is independent of the treatment assignment to the bin condition. That is evidence that the randomization procedure avoided any interference between the 2 treatments through the assignment process. Because there is no correlation with the bin treatment, inclusion of the SMS variable does not meaningfully change the confidence interval for the bin treatment effect.

It appears the authors use standard errors in their reporting rather than heteroskedastic consistent errors. The coefficients of this model match the reported coefficients in Table 4A, but the errors differ due to the treatment of heteroskedasticity.

### 3.7 Reproduce Table 4B, Column (2)

Now reproduce the results of column 2 in Table 4B, estimating separate treatment effects for the two types of SMS treatments and the two types of recycling-bin treatments. Provide a 95% confidence interval for the effect of the unadorned recycling bin. Explain how your answer differs from that in question 3.6, and explain why you think it differs.

```
mod_full <- lm(avg_bins_treat ~ bin_s + bin_g + sms_p + sms_g + no_cell_phone + base_avg_bins_treat + s

mod_full_rse <- sqrt(diag(vcovHC(mod_full)))
```

Table 7:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Any bin | 0.115*** |
| | (0.019) |
| Any SMS | 0.005 |
| | (0.024) |
| No cell phone | −0.047** |
| | (0.023) |
| Pre-treat Baseline | 0.373*** |
| | (0.030) |
| Constant | 0.385*** |
| | (0.038) |
| Street Fixed Effects | Yes |
| Observations | 1,781 |
| $R^2$ | 0.439 |
| Adjusted $R^2$ | 0.375 |
| Residual Std. Error | 0.323 (df = 1597) |
| F Statistic | 6.834*** (df = 183; 1597) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

```r
mod_full_ci <- ci_95(mod_full, 'bin_g')

mod_full_coef <- mod_full_ci$coef["bin_g"]
mod_full_coef_rse <- mod_full_ci$se['bin_g']
mod_full_lower <- mod_full_ci$lower['bin_g']
mod_full_upper <- mod_full_ci$upper['bin_g']


stargazer(mod_full,
          se = list(mod_full_rse),
          omit = "street",
          omit.labels = "Street Fixed Effects",
          covariate.labels = c("Bin with sticker", "Bin without sticker ",
                               "Personal SMS message","Generic SMS message",
                               "No cell phone", "Pre-treat Baseline", "Constant"),
          dep.var.labels = c("Avg \\# Visits Turned In"),
          type='latex', header=FALSE)
```

Table 8:

| | *Dependent variable:* |
|---|---|
| | Avg # Visits Turned In |
| Bin with sticker | 0.128*** |
| | (0.024) |
| Bin without sticker | 0.103*** |
| | (0.025) |
| Personal SMS message | −0.008 |
| | (0.028) |
| Generic SMS message | 0.020 |
| | (0.028) |
| No cell phone | −0.046** |
| | (0.023) |
| Pre-treat Baseline | 0.374*** |
| | (0.030) |
| Constant | 0.385*** |
| | (0.038) |
| Street Fixed Effects | Yes |
| Observations | 1,781 |
| $R^2$ | 0.440 |
| Adjusted $R^2$ | 0.375 |
| Residual Std. Error | 0.323 (df = 1595) |
| F Statistic | 6.769*** (df = 185; 1595) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

**Answer:** The coefficient for the Bin without sticker treatment effect is 0.103 with a heteroskedastic consistent standard error of 0.025. This is significant at the 1% level with a 95% confidence interval of [0.054, 0.152].

This model splits the Any bin treatment into 2 groups based on the sticker status of the bin. The coefficient that as being measured in previous model is now split into 2 groups based on a treatment condition. It is not unexpected to see a change in the coefficient and standard errors here. There are fewer participants in this group compared to the previous Any bin variable which combined Bin with stick and Bin without sticker, so the standard error is higher going from 0.019 for Any bin to 0.025 for Bin without sticker. The coefficient is lower because Any bin was essentially a weighted average of the 2 new coefficients for the sticker status. By splitting it into coefficients measuring with and without sticker separately, we can see that Bin with sticker has a slightly larger magnitude of effect.

Again, the authors use standard errors in their reporting rather than heteroskedastic consistent errors. The coefficients of this model match the reported coefficients in Table 4B, but the errors differ due to the treatment of heteroskedasticity.

# 4 A Final Practice Problem

Now for a fictional scenario. An emergency two-week randomized controlled trial of the experimental drug ZMapp is conducted to treat Ebola. (The control represents the usual standard of care for patients identified with Ebola, while the treatment is the usual standard of care plus the drug.)

Here are the (fake) data.

You are asked to analyze it. Each patient has their temperature and whether they are dehydrated recorded on day 0 of the experiment, then ZMapp is administered to patients in the treatment group on day 1. Dehydration and temperature is again recorded on day 14.

## 4.1 Simple treatment effect of Zmapp

Without using any covariates, answer this question with regression: What is the estimated effect of ZMapp (with standard error in parentheses) on whether someone was dehydrated on day 14? What is the p-value associated with this estimate?

```
zmapp_1 <- lm(dehydrated_day14 ~ treat_zmapp, data=d)

zmapp_1_results <- lm_results(zmapp_1, 'treat_zmapp')

zmapp_1_se <- zmapp_1_results['model_se']
zmapp_1_coef <- zmapp_1_results['zmapp_coef']
zmapp_1_coef_se <- zmapp_1_results['zmapp_coef_se']
zmapp_1_coef_p <- zmapp_1_results['zmapp_coef_p']


stargazer(zmapp_1,
          se = zmapp_1_se,
          covariate.labels = c("Treated with ZMapp"),
          dep.var.labels = c("Dehydrated on Day 14"),
          type='latex', header=FALSE)
```

Table 9:

| | *Dependent variable:* |
|---|---|
| | Dehydrated on Day 14 |
| Treated with ZMapp | $-0.238^{***}$ |
| | (0.091) |
| Constant | $0.847^{***}$ |
| | (0.048) |
| Observations | 100 |
| $R^2$ | 0.073 |
| Adjusted $R^2$ | 0.063 |
| Residual Std. Error | 0.421 (df = 98) |
| F Statistic | $7.705^{***}$ (df = 1; 98) |
| *Note:* | $^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01 |

**Answer:**

The estimated effect of Treated with ZMapp is c(treat_zmapp = -0.238) (c(treat_zmapp = 0.091)). Using

heterosketastic consistent standard errors this is significant at the 1% level (p = 0.007)

## 4.2 Add baseline covariates

Add covariates for dehydration on day 0 and patient temperature on day 0 to the regression from part (a) and report the ATE (with standard error). Also report the p-value.

```
zmapp_2 <-  lm(dehydrated_day14 ~ treat_zmapp + dehydrated_day0 +
                 temperature_day0, data=d)

zmapp_2_results <- lm_results(zmapp_2, 'treat_zmapp')

zmapp_2_se <- zmapp_2_results['model_se']
zmapp_2_coef <- zmapp_2_results['zmapp_coef']
zmapp_2_coef_se <- zmapp_2_results['zmapp_coef_se']
zmapp_2_coef_p <- zmapp_2_results['zmapp_coef_p']

stargazer(zmapp_2,
          se = zmapp_2_se,
          covariate.labels = c("Treated with ZMapp", "Dehydrated on Day 0",
                               "Temperature on Day 0"),
          dep.var.labels = c("Dehydrated on Day 14"),
          type='latex', header=FALSE)
```

Table 10:

| | *Dependent variable:* |
|---|---|
| | Dehydrated on Day 14 |
| Treated with ZMapp | $-0.166^{**}$ |
| | (0.082) |
| Dehydrated on Day 0 | 0.065 |
| | (0.178) |
| Temperature on Day 0 | $0.206^{***}$ |
| | (0.078) |
| Constant | $-19.470^{**}$ |
| | (7.608) |
| Observations | 100 |
| $R^2$ | 0.311 |
| Adjusted $R^2$ | 0.290 |
| Residual Std. Error | 0.367 (df = 96) |
| F Statistic | $14.447^{***}$ (df = 3; 96) |
| *Note:* | $^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01 |

**Answer:**

The estimated effect of Treated with ZMapp is -0.166 (0.082) when including covariates for temperature on day 0 and dehydrated on day 0. Using heterosketastic consistent standard errors this is significant at the 5% level (p = 0.031)

## 4.3 Interpret estimates

Do you prefer the estimate of the ATE reported in the chunk called `dehydration model` or `add pre-treatment measures`? Why? Report the results of the F-test that you used to form this opinion.

```
zmapp_test_object <- anova(zmapp_1, zmapp_2, test="F")

test_zm_f <- zmapp_test_object$F[2]
test_zm_rss_1 <- round(zmapp_test_object$`RSS`[1],3)
test_zm_rss_2 <- round(zmapp_test_object$`RSS`[2],3)
test_zm_p <- sprintf("%.4f", zmapp_test_object$`Pr(>F)`[2])

zmapp_test_object
```

```
## Analysis of Variance Table
##
## Model 1: dehydrated_day14 ~ treat_zmapp
## Model 2: dehydrated_day14 ~ treat_zmapp + dehydrated_day0 + temperature_day0
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1     98 17.383
## 2     96 12.918  2    4.4653 16.592 6.472e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Answer:**

I prefer the pre-treatment measures model because it accounts for the pre-treatment condition of the patients, which is correlated with their dehydrated status 14 days later. The F-test confirms the improvement is significant: F = 16.592, p = 0.0000 (<0.1%), with RSS falling from 17.383 to 12.918.

## 4.4 Add day fourteen temperature

The regression from part `add pre-treatment measures` suggests that temperature is highly predictive of dehydration. Add, temperature on day 14 as a covariate and report the ATE, the standard error, and the p-value.

```
zmapp_3 <-  lm(dehydrated_day14 ~ treat_zmapp + temperature_day14 +
                 dehydrated_day0 + temperature_day0 , data=d)

zmapp_3_results <- lm_results(zmapp_3, 'treat_zmapp')

zmapp_3_se <- zmapp_3_results['model_se']
zmapp_3_coef <- zmapp_3_results['zmapp_coef']
zmapp_3_coef_se <- zmapp_3_results['zmapp_coef_se']
zmapp_3_coef_p <- zmapp_3_results['zmapp_coef_p']

stargazer(zmapp_3,
          se = zmapp_3_se,
          covariate.labels = c("Treated with ZMapp", "Temperature on Day 14",
                               "Dehydrated on Day 0", "Temperature on Day 0"),
          dep.var.labels = c("Dehydrated on Day 14"),
          type='latex', header=FALSE)
```

**Answer:**

The estimated effect of Treated with ZMapp is -0.12 (0.086) when including covariates for temperature on day 14, temperature on day 0, and dehydrated on day 0. Using heterosketastic consistent standard errors this is not significant (p = 0.125).

Table 11:

| | *Dependent variable:* |
|---|---|
| | Dehydrated on Day 14 |
| Treated with ZMapp | $-0.120$ |
| | (0.086) |
| Temperature on Day 14 | $0.060^{**}$ |
| | (0.026) |
| Dehydrated on Day 0 | 0.046 |
| | (0.173) |
| Temperature on Day 0 | $0.177^{**}$ |
| | (0.077) |
| Constant | $-22.592^{***}$ |
| | (7.746) |
| Observations | 100 |
| $R^2$ | 0.340 |
| Adjusted $R^2$ | 0.312 |
| Residual Std. Error | 0.361 (df = 95) |
| F Statistic | $12.244^{***}$ (df = 4; 95) |
| *Note:* | $^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01 |

## 4.5 Interpret estimates

Do you prefer the estimate of the ATE reported in part `add pre-treatment measures` or `add day 14 temperature`? What is this preference based on?

**Answer:**

I prefer the model with just pre-treatment measures because temperature on day 14 is a bad control. The temperature on day 14 is a post-treatment covariate that may also be affected by the Zmapp treatment. Post-treatment covariates could block the model from properly attributing that variation in dehydration on day 14 caused by Zmapp, biasing the estimate toward zero.

## 4.6 Look at temperature

Now let's switch from the outcome of dehydration to the outcome of temperature, and use the same regression covariates as in the chunk titled `add pre-treatment measures`. Test the hypothesis that ZMapp is especially likely to reduce mens' temperatures, as compared to womens', and describe how you did so. What do the results suggest?

```
zmapp_4 <-  lm(temperature_day14 ~ treat_zmapp + dehydrated_day0 +
                 temperature_day0 + male + treat_zmapp * male, data=d)

zmapp_4_se <- sqrt(diag(vcovHC(zmapp_4)))


zmapp_4_coef_zmapp <- summary(zmapp_4)$coefficients[ 'treat_zmapp','Estimate']

zmapp_4_coef_male <- summary(zmapp_4)$coefficients[ 'male','Estimate']

zmapp_4_coef_malexzmapp <- summary(zmapp_4)$coefficients[ 'treat_zmapp:male','Estimate']
zmapp_4_coef_malexzmapp_p <- sprintf("%.4f",summary(zmapp_4)$coefficients[ 'treat_zmapp:male','Pr(>|t|)


stargazer(zmapp_4,
          se = list(zmapp_4_se),
          covariate.labels = c("Treated with ZMapp","Dehydrated on Day 0",
                               "Temperature on Day 0", "Male",
                               "Treated with ZMapp x Male"),
          dep.var.labels = c("Temperature on Day 14"),
          type='latex', header=FALSE)
```

**Answer:**

To test the hypothesis that ZMapp has a heterogeneous treatment effect on day 14 temperature by gender, I added the covariate for male and the interaction between male and the ZMapp treatment to the regression model. Without ZMapp, men have a temperature 3.085 degrees higher than women. The ZMapp treatment reduced temperature for women by -0.231 degrees. The Male x Zmapp interaction term is -2.077 ($p = 0.0000$), which is a statistically significant difference in treatment effect. The total treatment effect for men is -0.231 + -2.077 = -2.308. This supports the hypothesis that ZMapp has a heterogeneous treatment effect on men's and women's temperatures.

## 4.7 Compare health outcomes

Which group – those that are coded as `male == 0` or `male == 1` have better health outcomes (temperature) in control? What about in treatment? How does this help to contextualize whatever heterogeneous treatment effect you might have estimated?

Table 12:

| | *Dependent variable:* |
|---|---|
| | Temperature on Day 14 |
| Treated with ZMapp | $-0.231^{*}$ |
| | (0.118) |
| Dehydrated on Day 0 | 0.041 |
| | (0.195) |
| Temperature on Day 0 | $0.505^{***}$ |
| | (0.105) |
| Male | $3.085^{***}$ |
| | (0.122) |
| Treated with ZMapp x Male | $-2.077^{***}$ |
| | (0.198) |
| Constant | $48.713^{***}$ |
| | (10.194) |
| Observations | 100 |
| $R^2$ | 0.906 |
| Adjusted $R^2$ | 0.901 |
| Residual Std. Error | 0.452 (df = 94) |
| F Statistic | $180.953^{***}$ (df = 5; 94) |
| *Note:* | $^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01 |

```r
temps_matrix <- d[, .(mean_temp = mean(temperature_day14)), keyby = .(treat_zmapp, male)]

control_w <- round(temps_matrix$mean_temp[1],2)
control_m <- round(temps_matrix$mean_temp[2],2)
treat_w <- round(temps_matrix$mean_temp[3],2)
treat_m <- round(temps_matrix$mean_temp[4],2)

temps_matrix
```

```
## Key: <treat_zmapp, male>
##    treat_zmapp  male mean_temp
##          <int> <int>     <num>
## 1:           0     0  98.48654
## 2:           0     1 101.69167
## 3:           1     0  98.16308
## 4:           1     1  99.10071
```

**Answer:**

In control, women have a lower day 14 temperature than men, 98.49 compared to 101.69. After treatment, women's average temperature fell slightly to 98.16, and men's average temperature fell -2.59 degrees to 99.1. The treatment effect is stronger in `male == 1` because Ebola has a different effect on men's and women's temperatures in this study.

## 4.8 Collaborating with others, Part (1)

Suppose you speak with a colleague to learn about heterogeneous treatment effects.

This colleague has access to a non-anonymized version of the same dataset and reports that they looked at heterogeneous effects of the ZMapp treatment by each of 80 different covariates to examine whether each predicted the effectiveness of ZMapp on each of 20 different indicators of health.

Across these regressions your colleague ran, the treatment's interaction with sex on the outcome of temperature is the only heterogeneous treatment effect that he found to be statistically significant. They reason that this shows the importance of sex for understanding the effectiveness of the drug, because nothing else seemed to indicate why it worked. Bolstering your colleague's confidence, after looking at the data, they also returned to their medical textbooks see the whispers of a theory about why ZMapp interacts with processes only present in men to cure.

Another doctor, unfamiliar with the data, hears your colleague's theory and finds it plausible. How likely do you think it is ZMapp works especially well for curing Ebola in men, and why? (This question is conceptual can be answered without performing any computation.)

**Answer:**

Our colleague has performed 1,600 tests (80 covariates x 20 outcomes), and only 1 of those tests (ZMapp x Sex) showed a heterogeneous treatment effect that is statistically significant. The high volume of tests conducted is a red flag in this situation and we should be cautious to draw any conclusions about significant results. With that many tests using $\alpha = 0.05$, there should be some that register as statistically significant on pure random chance. The fact that only 1 is significant could be a sign of a testing procedure error, as we would expect 1,600 x 0.05 = 80 tests to be significant on average. I would be very suspect of this result until it has been more thoroughly tested, ideally in a separate follow-up experiment.

With respect to the temperature and sex interaction specifically, our colleague may be missing the full story in the data. In the control group, men have an elevated temperature compared to women that is statistically significant. This is really the big question of the study: why do the women who were untreated exhibit a normal temperature on average while men exhibit a fever under the same conditions? The colleague is looking at the change in temperature between men in control and men in treatment, but they need to look at

the difference in temperature for men in control and women in control before they can make the claim that ZMapp has a heterogeneous treatment effect.

## 4.9 Collaborating with others, Part (2)

Suppose that your colleague conducted their research looking at the interaction of 80 covariates with ZMapp, but that you on your own tested this and only this HTE, and discovered a positive result. How, if at all, does your colleague's behavior change the interpretation of your test? Does this seem fair or reasonable?

**Answer:**

The prior knowledge in this situation has a big impact on the interpretation of results.

If we have prior knowledge that 1,599 tests are not worth executing, but we go and verify the result of the 1 test that is significant, we cannot draw a major conclusion about the heterogeneous treatment effect from that. To use an analogy, our colleague found the needle in the haystack, and told us exactly where to find it. The evidence for a heterogeneous treatment effect is weak, and our 1 test does nothing to strengthen the case.

If we run one and only one test with no prior knowledge of the 1,600 results, that does change the interpretation. For any single test we have the same $\alpha = 0.05$ chance of a false positive, and we should consider this a valid result. At this point, we have no reason to be more suspicious of this result than any other individual test we run on other data sets.

After we find our significant result and then go and share knowledge with our colleague who ran 1,600 tests, we are at the same situation as the prior knowledge before we run the tests. We now have knowledge and expectation of the results for the remaining covariates and outcomes. It is no longer reasonable to rerun the tests knowing how they will turn out (except if we believe the tests were not run correctly, and these results are not trustworthy). If we really want to build our understanding of the heterogeneous treatment effect of ZMapp and sex, we should run a new study that is designed to test that result.

## 4.10 Collaborating with others, Part (3)

Now, imagine that your colleague had not conducted the 80 different regressions. Instead, they tested this heterogeneous treatment effect, and only this heterogeneous treatment effect, of their own accord. Would you be more or less inclined to believe that the heterogeneous treatment effect really exists? Why? Is there a general principle that is guiding your reasoning?

**Answer:**

In general, we have to consider the prior knowledge of the situation no matter who is running the test. Whether it is us or our colleague who runs the test is less important than what other tests were run prior to this result. The statistical tools we use are designed to give us a result with a reasonable false positive rate determined by $\alpha$. A single test run in good faith with no prior knowledge of other test results should be viewed as valid and fair until we have evidence to the contrary. I would be inclined to believe the single test result.

A natural follow up question to this result would be asking if the heterogeneous treatment effect is exhibited in other outcomes or covariates. Immediately running 1,600 more tests still is not the right next move. The more tests we run, the greater the risk that we encounter at least 1 false positive along the way. The next step should be an intentional and good faith effort to verify the HTE. There may be other variables in our data set that make sense to test given our understanding of ZMapp, Ebola, and gender. If this is a question of critical importance then it may require a separate experiment to verify the existence and significance of the effect.